

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

● ANSWER KEY

Geometry and Trigonometry

03

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1**B****B.** 30°

Choice B is correct. It's given that quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P , Q , R , and S correspond to P' , Q' , R' , and S' , respectively. Since corresponding angles of similar quadrilaterals are congruent, it follows that the measure of angle P is equal to the measure of angle P' . It's given that the measure of angle P is 30° . Therefore, the measure of angle P' is 30° .

Choice A is incorrect. This is $\frac{1}{3}$ the measure of angle P' .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is 3 times the measure of angle P' .

Q2**D****D.** $94 \cdot 94$

Choice D is correct. The area of a square is given by s^2 , where s is the length of each side of the square. It's given that the length of each side of a square is 94 cm. It follows that the area, in cm^2 , of the square is 94^2 , or $94 \cdot 94$. Therefore, the expression that gives the area, in cm^2 , of the square is $94 \cdot 94$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This expression gives the perimeter, in cm, of the square.

Q3**A**

A. The center is at $(13, k)$ and the radius is 8.

Choice A is correct. For a circle in the xy -plane that has the equation $x - h^2 + y - k^2 = r^2$, where h , k , and r are constants, h, k is the center of the circle and the positive value of r is the radius of the circle. In the given equation, $h = 13$ and $r^2 = 64$. Taking the square root of each side of $r^2 = 64$ yields $r = \pm 8$. Therefore, the center of the circle is at $13, k$ and the radius is 8.

Choice B is incorrect. This gives the center and radius of a circle with equation $x - k^2 + y - 13^2 = 64$, not $x - 13^2 + y - k^2 = 64$.

Choice C is incorrect. This gives the center and radius of a circle with equation $x - k^2 + y - 13^2 = 4,096$, not $x - 13^2 + y - k^2 = 64$.

Choice D is incorrect. This gives the center and radius of a circle with equation $x - 13^2 + y - k^2 = 4,096$, not $x - 13^2 + y - k^2 = 64$.

Q4**B**

B. $\frac{308}{317}$

Choice B is correct. If two triangles are similar, then their corresponding angles are congruent. It's given that right triangle FGH is similar to right triangle JKL and angle F corresponds to angle J . It follows that angle F is congruent to angle J and, therefore, the measure of angle F is equal to the measure of angle J . The sine ratios of angles of equal measure are equal. Since the measure of angle F is equal to the measure of angle J , $\sin F = \sin J$. It's given that $\sin F = \frac{308}{317}$. Therefore, $\sin J$ is $\frac{308}{317}$.

Choice A is incorrect. This is the value of $\cos J$, not the value of $\sin J$.

Choice C is incorrect. This is the reciprocal of the value of $\sin J$, not the value of $\sin J$.

Choice D is incorrect. This is the reciprocal of the value of $\cos J$, not the value of $\sin J$.

Q5**84**

The correct answer is 84. The sum of the measures of the interior angles of a triangle is 180° . It's given that in triangle JKL , the measures of $\angle K$ and $\angle L$ are each 48° . Adding the measures, in degrees, of $\angle K$ and $\angle L$ gives $48 + 48$, or 96. Therefore, the measure of $\angle J$, in degrees, is $180 - 96$, or 84.

Q6**C****C.** 726π

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of 6 centimeters. Therefore, $h = 6$. It's also given that the cylinder has a base diameter of 22 centimeters. The radius of a circle is half the diameter of the circle. Since the base of a right circular cylinder is a circle, it follows that the radius of the base of the right circular cylinder is $\frac{22}{2}$, or 11, centimeters. Therefore, $r = 11$. Substituting 11 for r and 6 for h in the formula $V = \pi r^2 h$ yields $V = \pi 11^2 6$, which is equivalent to $V = \pi 121 6$, or $V = 726\pi$. Therefore, the volume, in cubic centimeters, of the cylinder is 726π .

Choice A is incorrect. This is the volume of a right circular cylinder that has a base diameter of $2\sqrt{22}$, not 22, centimeters and a height of 6 centimeters.

Choice B is incorrect. This is the volume of a right circular cylinder that has a base diameter of $4\sqrt{11}$, not 22, centimeters and a height of 6 centimeters.

Choice D is incorrect. This is the volume of a right circular cylinder that has a base diameter of 44, not 22, centimeters and a height of 6 centimeters.

Q7**D**

D. $x + 4^2 + y - 5^2 = 25$

Choice D is correct. A circle in the xy -plane can be represented by an equation of the form $x - h^2 + y - k^2 = r^2$, where h, k is the center of the circle and r is the length of a radius of the circle. It's given that the circle has its center at $-4, 5$. Therefore, $h = -4$ and $k = 5$. Substituting -4 for h and 5 for k in the equation $x - h^2 + y - k^2 = r^2$ yields $x - 4^2 + y - 5^2 = r^2$, or $x + 4^2 + y - 5^2 = r^2$. It's also given that the point $-8, 8$ lies on the circle. Substituting -8 for x and 8 for y in the equation $x + 4^2 + y - 5^2 = r^2$ yields $-8 + 4^2 + 8 - 5^2 = r^2$, or $-4^2 + 3^2 = r^2$, which is equivalent to $16 + 9 = r^2$, or $25 = r^2$. Substituting 25 for r^2 in the equation $x + 4^2 + y - 5^2 = r^2$ yields $x + 4^2 + y - 5^2 = 25$. Thus, the equation $x + 4^2 + y - 5^2 = 25$ represents the circle.

Choice A is incorrect. The circle represented by this equation has its center at $4, -5$, not $-4, 5$, and the point $-8, 8$ doesn't lie on the circle.

Choice B is incorrect. The point $-8, 8$ doesn't lie on the circle represented by this equation.

Choice C is incorrect. The circle represented by this equation has its center at $4, -5$, not $-4, 5$, and the point $-8, 8$ doesn't lie on the circle.

Q8**4**

The correct answer is 4. Triangle PQR has given angle measures of 30° and 90° , so the third angle must be 60° because the measures of the angles of a triangle sum to 180° . For any special right triangle with angles measuring 30° , 60° , and 90° , the length of the hypotenuse (the side opposite the right angle) is $2x$, where x is the length of the side opposite the 30° angle. Segment PQ is opposite the 30° angle. Therefore, $2(PQ) = 8$ and $PQ = 4$.

Q9**D****D.** 12

Choice D is correct. It's given that triangle ABC is similar to triangle XYZ , where A , B , and C correspond to X , Y , and Z , respectively. It follows that side AB corresponds to side XY and side BC corresponds to side YZ . Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{XY}{AB} = \frac{YZ}{BC}$. Substituting 170 for AB , 60 for YZ , and 850 for BC in this equation yields $\frac{XY}{170} = \frac{60}{850}$. Multiplying each side of this equation by 170 yields $XY = 12$. Therefore, the length of $\overline{XY}$ is 12.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length of $\overline{YZ}$, not $\overline{XY}$.

Q10**D****D.** 68,921

Choice D is correct. The volume, V , of a cube can be found using the formula $V = s^3$, where s is the edge length of the cube. It's given that a cube has an edge length of 41 inches. Substituting 41 inches for s in this equation yields $V = 41^3$ cubic inches, or $V = 68,921$ cubic inches. Therefore, the volume of the cube is 68,921 cubic inches.

Choice A is incorrect. This is the perimeter, in inches, of the cube.

Choice B is incorrect. This is the area, in square inches, of a face of the cube.

Choice C is incorrect. This is the surface area, in square inches, of the cube.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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